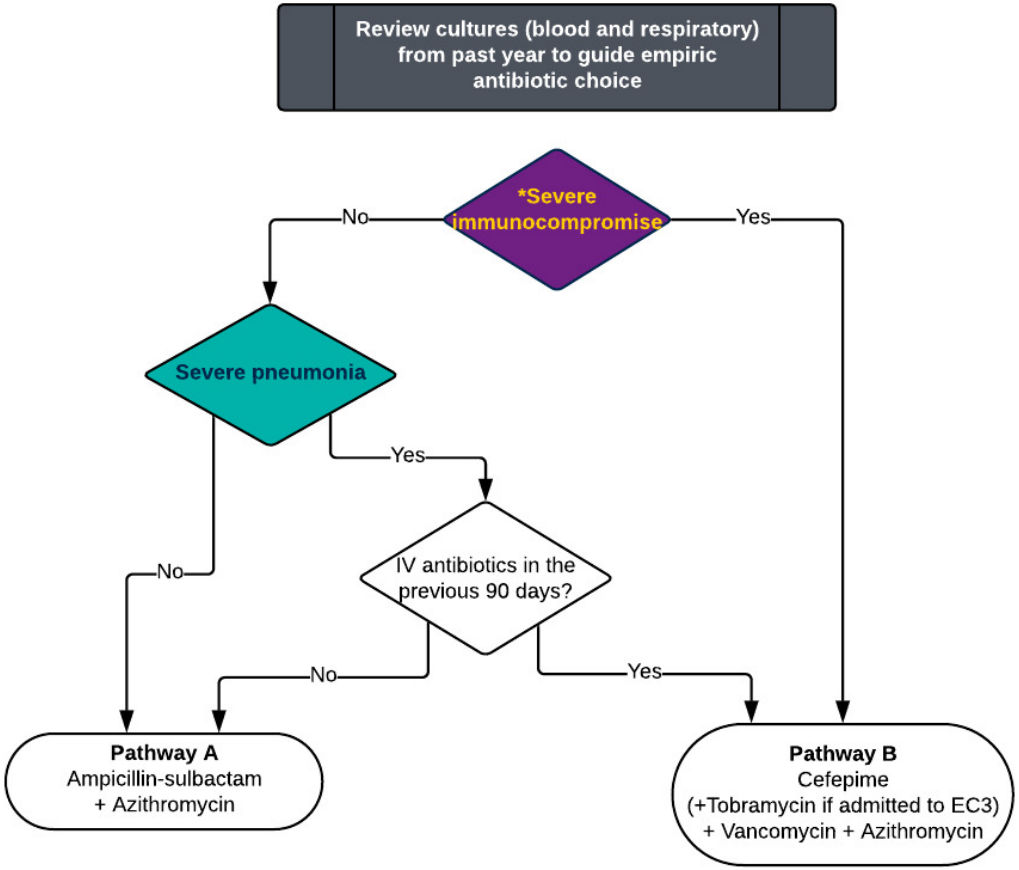
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Treatment Pathway for Adult Patients with Pneumonia

The purpose of this document is to guide the appropriate treatment of adult patients presenting with pneumonia. Empiric treatment regimens based on risk of infection with multidrug-resistant (MDR) pathogens (including MRSA, Pseudomonas spp., Acinetobacter spp., organisms not susceptible to beta-lactams (ceftriaxone or ampicillin-sulbactam) and/or fluoroquinolones (ciprofloxacin, levofloxacin)) are shown below.

Pathway A	Pathway B	Pathway C	Pathway D
Non-ICU CAP	Non-ICU CAP with MDRO risk factors	Non-ICU HAP	ICU CAP/HAP/VAP

Patient in AES



Community-acquired:
This includes all patients presenting from the community, including nursing homes, dialysis patients, LTCF, recent hospitalization

Severe immunocompromise:
- AIDS (CD4 < 200)
- Neutropenia (ANC < 1000)
- Lung Transplant
- Solid organ transplant < 1 year or rejection < 6 months
- Allogeneic stem cell transplant < 1 year or with GvHD

*certain **Moderate immunocompromised** patients can get targeted CAP therapy (Pathway A) if they are admitted to the floor (non-severe). This includes:
- Solid tumor on chemotherapy
- Hematologic malignancy
- Solid organ transplant (non-lung) > 1 year
- Asplenia
- Patients receiving immunosuppressive medications (e.g., TNF-a inhibitors, rituximab, steroids, methotrexate)

Severe pneumonia:
This includes all patients presenting with septic shock or requiring EC3

Pathway A			
Indication	**Empiric Therapy**	**Duration**	**Comments**
Inpatient, non-ICU community -acquired pneumonia, no MDRO risk factors	Preferred: **Ampicillin-sulbactam*** 3 g IV q6h **+ Azithromycin** 500 mg IV/PO x1 day, then 250 mg q24h x4 days Low/high risk PCN allergy***: **Ceftriaxone** 2 g IV q24h **+ Azithromycin** 500 mg IV/PO x1 day, then 250 mg q24h x4 days Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO q12h *if* empyema or lung abscess present Severe non-IgE PCN allergy***: **Levofloxacin*** 750 mg IV/PO q24h Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO q12h *if* empyema or lung abscess present Addition of vancomycin Consider in the setting of post-influenza pneumonia	Uncomplicated CAP: • **5 days** for patients who defervesce within 72 hours and have *no more than* 1 sign of CAP instability† at the time of antibiotic discontinuation. • Patients with delayed response should discontinue therapy 48-72 hours after defervescing and have *no more than* 1 sign of CAP instability† at time antibiotic discontinuation. • Aspiration pneumonia should receive same duration as uncomplicated pneumonia (see Comments). Complicated Pneumonia: • Patients with empyema, infected pleural effusions, and bacteremia secondary to pneumonia may require longer durations of therapy. Bacteremic pneumococcal pneumonia should be treated for a minimum of 7-14 days. ID consult is recommended for patients with complicated pneumonia. Pathogen-specific considerations: • Uncomplicated pneumonia with non-fermenting GNRs (e.g., *Pseudomonas, Achromobacter, Acinetobacter, Stenotrophomonas*) or *Staphylococcus aureus* should receive 7 days of therapy if defervescence within 72 hours and have *no more than 1* sign of CAP instability at the time of antibiotic discontinuation†. Delayed response will likely require longer durations.	Aspiration: Patients with known aspiration may have pneumonitis with fever, leukocytosis, new infiltrates and hypoxia within the first 24 hours of the event, which is prior to when bacterial pneumonia would be expected. Antimicrobials are not indicated for pneumonitis and do not prevent development of pneumonia. Anaerobic coverage is not necessary for patients with pneumonia following an aspiration event. Only those with empyema or lung abscess should receive empiric anaerobic coverage. For culture negative pneumonia, transition to oral therapy when patient is afebrile with clinical improvement and hemodynamically stable for 48 hours: • 1st line: **Amoxicillin-clavulanate*** 875 mg BID **+ Azithromycin** (complete 5-day course of azithromycin) • Low/high risk PCN allergy***: **Cefuroxime*** 500 mg BID **+ Azithromycin** (complete 5-day course of azithromycin) • Severe non-IgE mediated PCN allergy***: **Levofloxacin*** 750 mg PO q24h Pathogen-specific recommendations: Use **azithromycin** 500 mg q24h *if documented or high clinical suspicion for Legionella* (can pursue further diagnostic testing → respiratory legionella PCR). In setting of macrolide allergy can use **doxycycline** for atypical coverage in absence of *Legionella* concern. In patients with documented Mycoplasma, use of doxycycline should be preferred for treatment due to concern for macrolide resistance. Antibiotic coverage of atypical organisms can be discontinued if the respiratory panel (RPAN) and *Legionella* urine antigen are negative and suspicion is low. In the setting of a respiratory panel positive for a viral pathogen that could explain the clinical presentation (not rhino/enterovirus), and when the clinical history and symptoms, radiographic findings, and lab results do not support a superimposed bacterial infection, antibiotic discontinuation is appropriate. Tips on *utilization of procalcitonin (PCT) levels*

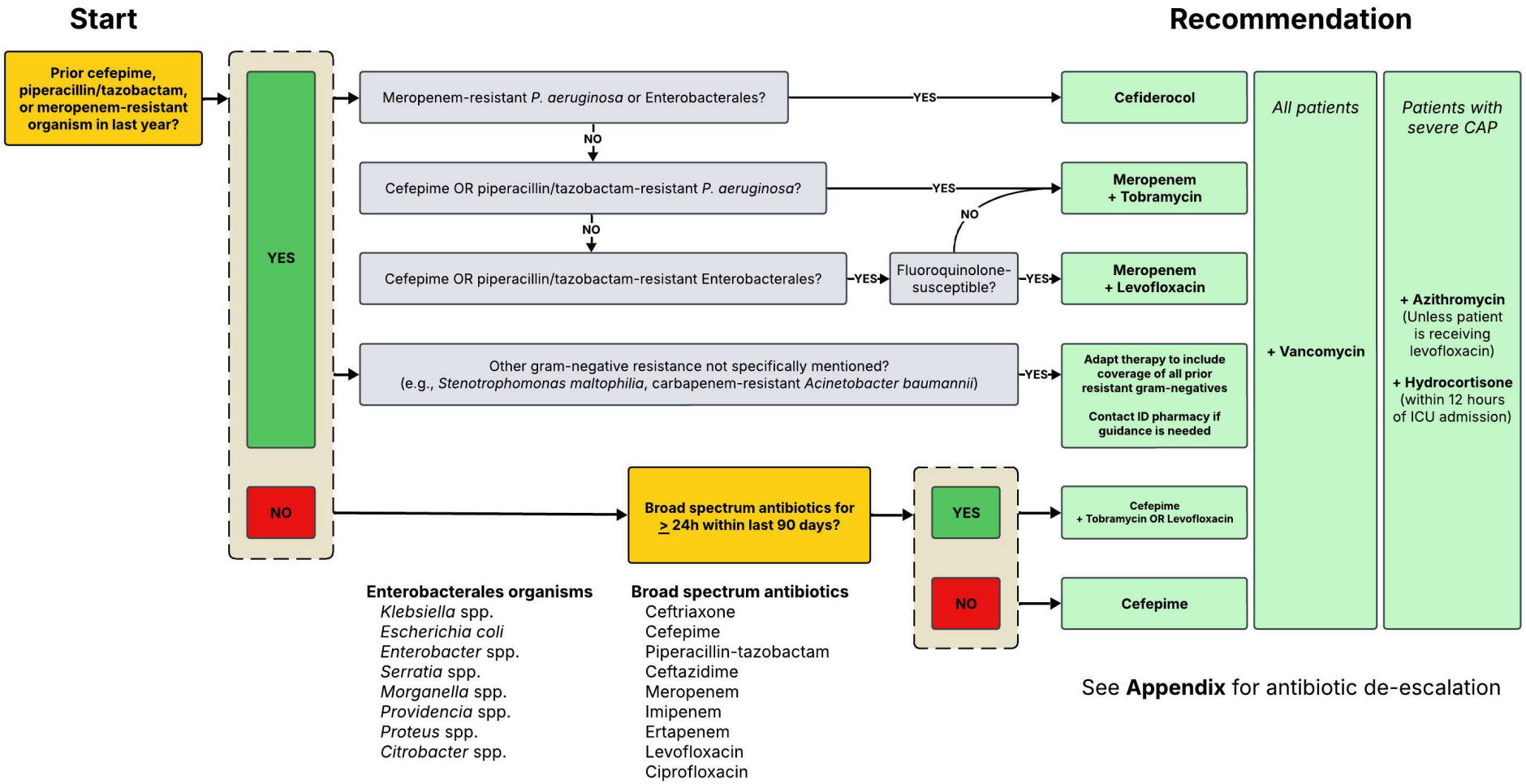
Pathway B
(Previous culture data should be used to guide empiric therapy)

Indication	Empiric Therapy	Duration	Comments
Inpatient, non-ICU community-acquired pneumonia with risk factors for MDRO (below) **Patients presenting with any of the following risk factors for drug-resistant pathogens:** ● History of infection or colonization with Pseudomonas spp., MRSA, or other MDR gram-negative pathogens (resistant to ampicillin-sulbactam or ceftriaxone) within previous 12 months **OR** ● In patients with severe community-acquired pneumonia (septic shock or EC3) **AND prior hospitalization for at least 48 hours AND use of any intravenous antibiotic, fluoroquinolone, or linezolid within previous 90 days** **OR** ● Patients with severe immunocompromising condition, defined as: ○ AIDS (CD4 < 200) ○ Neutropenia (ANC < 1000) ○ Lung transplant recipient ○ Solid organ transplant recipient (non-lung) within previous 1 year ○ Solid organ transplant recipient treated for rejection within previous 6 months ○ Allogeneic stem cell transplant within previous 1 year or those with chronic GVHD	Preferred (safe to use in low-high risk PCN allergy ***): **Cefepime*** 2 g IV q8h (+ **Tobramycin*** IV if admitted to EC3) + **Vancomycin** IV (see nomogram) + **Azithromycin** 500 mg IV/PO Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO/IV q12h *if* empyema or lung abscess present NOTE: If patient has a history of Pseudomonas or MDR gram-negative pathogen ONLY, empiric vancomycin use is not necessary. If MRSA history ONLY, use of cefepime is not necessary (preferred regimen would be ampicillin-sulbactam + vancomycin). Low/high risk cephalosporin allergy***: **Meropenem*** 2 g IV q8h (+ **Tobramycin*** IV if admitted to EC3) + **Vancomycin** IV (see nomogram) + **Azithromycin** 500 mg IV/PO Severe non-IgE mediated PCN or cephalosporin allergy***: **Aztreonam*** 2 g IV q8h (+ **Tobramycin*** IV if admitted to EC3) + **Vancomycin** IV (see nomogram) + **Azithromycin** 500 mg IV/PO Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO/IV q12h *if* empyema or lung abscess present **Linezolid** may be used in patients with vancomycin allergy (not vancomycin infusion reaction). See restriction criteria for appropriate empiric and definitive use of linezolid.	Uncomplicated CAP: ● **5 days** for patients who defervesce within 72 hours and have *no more than* 1 sign of CAP instability† at the time of antibiotic discontinuation. ● Patients with delayed response should discontinue therapy 48-72 hours after defervesce and have *no more than* 1 sign of CAP instability† at time antibiotic discontinuation. ● Aspiration pneumonia should receive same duration as uncomplicated pneumonia (see Comments). Complicated Pneumonia: ● Patients with empyema, infected pleural effusions, and bacteremia secondary to pneumonia should receive longer durations of therapy. Bacteremic pneumococcal pneumonia should be treated for a minimum of 7-14 days. ID consult is recommended for patients with complicated pneumonia. Pathogen-specific considerations: ● Uncomplicated pneumonia with non-fermenting GNRs (e.g., *Pseudomonas, Achromobacter, Acinetobacter, Stenotrophomonas*) or *Staphylococcus aureus* should receive 7 days of therapy if defervescence within 72 hours and have *no more than 1* sign of CAP instability† at the time of antibiotic discontinuation. Delayed response will likely require longer durations. ● *Pseudomonas* pneumonia with complicating factors should be considered for 14-day duration (see pathway C for further clarification)	Aspiration: Patients with known aspiration may have pneumonitis with fever, leukocytosis, new infiltrates and hypoxia within the first 24 hours of the event, which is prior to when bacterial pneumonia would be expected. Antimicrobials are not indicated for pneumonitis and do not prevent development of pneumonia. *Definitive therapy should be tailored to culture results.* In patients with negative respiratory cultures who are clinically stable after 48 hours, de-escalate antibiotic therapy to CAP treatment. Oral antibiotics should be considered in clinically stable patients. ● Preferred: **Amoxicillin-clavulanate*** 875 mg BID ● Low/high risk PCN allergy***: **Cefuroxime*** 500 mg BID ● Severe non-IgE mediated PCN allergy***: **Levofloxacin*** 750 mg PO q24h Pathogen-specific recommendations: If atypical pathogens are suspected, start **azithromycin** 500 mg IV x1 day, followed by 250 mg IV/PO daily x4 days. Use **azithromycin** 500 mg q24h if high clinical suspicion for *Legionella*. Treatment duration may be longer for confirmed *Legionella*. In setting of macrolide allergy can use **doxycycline** for atypical coverage in non-ICU patients and in the absence of *Legionella* concern. Antibiotic coverage of atypical organisms can be discontinued if the respiratory panel (RPAN) and *Legionella* urine antigen are negative and suspicion is low. In patients with documented or high clinical suspicion for *Mycoplasma*, use of **doxycycline** should be preferred for treatment due to concern for macrolide resistance. Discontinue vancomycin if no evidence of MRSA colonization (negative MRSA nares PCR) if clinically appropriate. In patients with evidence of colonization (positive MRSA nares PCR) or if unknown, discontinue vancomycin after 48-72 hours if no positive respiratory cultures for MRSA and clinically appropriate. Vancomycin can be continued for gram-positive coverage in patients with no microbiological diagnosis who are receiving aztreonam due to allergies. Tips on *utilization of procalcitonin (PCT) levels*

Pathway C (Previous culture data should be used to guide empiric therapy)			
Indication	**Empiric Therapy**	**Duration**	**Comments**
Non-ICU, hospital-acquired pneumonia (current hospitalization for ≥ 72 hours)	Preferred (safe to use in low/high risk PCN allergy***): **Cefepime*** 2 g IV q8h + **Vancomycin** IV (see nomogram) Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO/IV q12h *if* empyema or lung abscess present Low/high risk cephalosporin allergy***: **Meropenem*** 2 g IV q8h + **Vancomycin** IV (see nomogram) Severe non-IgE mediated PCN and cephalosporin allergy***: **Aztreonam*** 2 g IV q8h + **Vancomycin** IV (see nomogram) Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO/IV q12h *if* empyema or lung abscess present **Linezolid** may be used in patients with vancomycin allergy (not vancomycin infusion reaction). See restriction criteria for appropriate empiric and definitive use of linezolid.	Uncomplicated Pneumonia: • **5 days** is appropriate for patients with a rapid clinical response within 72 hours without one of the pathogens listed below. • For non-fermenting GNRs (e.g., *Pseudomonas*, *Achromobacter*, *Acinetobacter*, *Stenotrophomonas*) or *Staphylococcus aureus*: **7 days** is recommended. • See below for considerations of longer durations in *Pseudomonas* pneumonia with complicating factors. Complicated Pneumonia: • Patients with empyema, infected pleural effusions, lung abscesses, and bacteremia secondary to pneumonia should receive longer durations of therapy. ID consult is recommended for patients with complicated pneumonia. *Pseudomonas* with complicating factors: 14 days *could be considered* for patients with *Pseudomonas* pneumonia AND any of the following: • Multiple recurrences of *Pseudomonas* pneumonia • Immunocompromised • Delayed clinical improvement > 3 days • Postobstructive pneumonia • Bronchiectasis	Antibiotic therapy is not generally recommended for patients with ventilator-associated tracheobronchitis (defined as fever with no other recognizable cause, with new or increased sputum production, positive endotracheal aspirate culture, and no radiographic evidence of pneumonia). Aspiration: Patients with known aspiration may have pneumonitis with fever, leukocytosis, new infiltrates and hypoxia within the first 24 hours of the event, which is prior to when bacterial pneumonia would be expected. Antimicrobials are not indicated for pneumonitis and do not prevent development of pneumonia. Consider 5 days of antibiotics if bacterial pneumonia is suspected. Pathogen-specific recommendations: Discontinue vancomycin if no evidence of MRSA colonization (negative MRSA nares PCR) if clinically appropriate. In patients with evidence of colonization (positive MRSA nares PCR) or if unknown, discontinue vancomycin after 48-72 hours if no positive respiratory cultures for MRSA and clinically appropriate. Vancomycin can be continued for gram-positive coverage in patients with no microbiological diagnosis who are receiving aztreonam due to allergies.

Pathway D			
Indication	**Empiric Therapy**	**Duration**	**Comments**
Pneumonia in ICU • Community-acquired pneumonia • Hospital-acquired pneumonia • Ventilator-associated pneumonia	See **Adult ICU Treatment Algorithm** *(page 6)* for recommendations Low/high risk cephalosporin allergy***: If cefepime is otherwise recommended, substitute **Meropenem*** 2 g IV q8h *Combination therapy should still be given to patients with prior broad-spectrum antibiotics if receiving meropenem due to allergies.* Severe non-IgE mediated PCN and cephalosporin allergy***: If cefepime or meropenem is otherwise recommended, substitute **Aztreonam*** 2 g IV q8h Consider the addition of anaerobic coverage with **metronidazole** 500 mg PO/IV q12h *if* empyema or lung abscess present **Linezolid** may be used in patients with vancomycin allergy (not vancomycin infusion reaction). See restriction criteria for appropriate empiric and definitive use of linezolid.	Uncomplicated Pneumonia: **7 days** is appropriate for patients with a rapid clinical response within 72 hours, regardless of pathogen (see below for considerations of longer durations in *Pseudomonas* pneumonia with complicating factors). Complicated Pneumonia: Patients with empyema, infected pleural effusions, lung abscesses, and bacteremia secondary to pneumonia should receive longer durations of therapy. ID consult is recommended for patients with complicated pneumonia. *Pseudomonas* with complicating factors: 14 days is *recommended* for patients with any of the following to decrease recurrence: • Circulatory shock • Acute respiratory distress syndrome • Extracorporeal membrane oxygenation (ECMO) 14 days *could be considered* for patients with *Pseudomonas* pneumonia AND any of the following: • Multiple recurrences of *Pseudomonas* pneumonia • Immunocompromised • Delayed clinical improvement > 3 days • Postobstructive pneumonia • Bronchiectasis	Antibiotic therapy is not generally recommended for patients with ventilator-associated tracheobronchitis (defined as fever with no other recognizable cause, with new or increased sputum production, positive endotracheal aspirate culture, and no radiographic evidence of pneumonia). The treatment algorithm does not directly account for all gram-negative resistant pathogens (see *"Other gram-negative resistance not specifically mentioned?"* in algorithm). If a patient has a history of a gram-negative organism resistant to the treatment pathway recommendation, adapt therapy to include coverage. Contact ID pharmacy at **35903** if further guidance with treatment selection is needed. **Tobramycin** is preferred for combination therapy in most patients. **Levofloxacin** may be substituted in cases where there is high clinical concern for toxicity. Severe community-acquired pneumonia: • Initiate **hydrocortisone** 50 mg IV q6h within 12 hours of ICU admission • Continue for a duration of 7-14 days based on clinical improvement OR discontinue when patient is no longer ICU status, whichever is shorter Pathogen-specific recommendations: A bronchoalveolar lavage (BAL) or mini-BAL, unless contraindicated, is strongly recommended within 24 hours of empiric antibiotic initiation to facilitate pathogen-directed treatment and early antibiotic de-escalation. Discontinue vancomycin if no evidence of MRSA colonization (negative MRSA nares PCR) if clinically appropriate. In patients with evidence of colonization (positive MRSA nares PCR) or if unknown, discontinue vancomycin after 48-72 hours if no positive respiratory cultures for MRSA and clinically appropriate. Cefiderocol and aztreonam are not active against gram-positive organisms, including MSSA. For patients receiving either cefiderocol or aztreonam, vancomycin should be continued for gram-positive coverage regardless of a negative MRSA screen until a microbiological diagnosis is confirmed. See **Appendix** *(page 7)* for detailed guidance on antibiotic de-escalation based on culture results. Tips on *utilization of procalcitonin (PCT) levels*

Patient in Adult ICU (Pathway D)



Enterobacterales organisms
- *Klebsiella* spp.
- *Escherichia coli*
- *Enterobacter* spp.
- *Serratia* spp.
- *Morganella* spp.
- *Providencia* spp.
- *Proteus* spp.
- *Citrobacter* spp.

Broad spectrum antibiotics
- Ceftriaxone
- Cefepime
- Piperacillin-tazobactam
- Ceftazidime
- Meropenem
- Imipenem
- Ertapenem
- Levofloxacin
- Ciprofloxacin

See **Appendix** for antibiotic de-escalation

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Appendix. Gram-negative Antibiotic De-escalation for Patients with Pneumonia in the ICU at 48h

Please re-evaluate if the patient has bacterial pneumonia and stop antibiotics if an alternative diagnosis is the cause of initial respiratory symptoms (e.g. viral infection, aspiration, COPD exacerbation, cardiac failure or shock, etc.)

Respiratory Culture Result

No Growth	Gram-Negative Growth	No Respiratory Culture Obtained
If using combination therapy, **stop** tobramycin or levofloxacin **No history of *P. aeruginosa* or ceftriaxone-resistant Enterobacterales:** • **Change** to ampicillin/sulbactam • Step-down to PO amoxicillin-clavulanate when clinically stable for 48-72h **History of *P. aeruginosa* or ceftriaxone-resistant Enterobacterales‡:** • Continue empiric beta-lactam to complete a 7-day course	**Organism ID**: • If using combination therapy, **stop** tobramycin or levofloxacin unless *P. aeruginosa* is identified on culture **Susceptibilities**: • *P. aeruginosa* ○ **Stop** tobramycin or levofloxacin once on beta-lactam with confirmed susceptibility • All other organisms ○ Tailor antimicrobial therapy accordingly, avoid definitive therapy with anti-pseudomonal if feasible	**Early clinical improvement:** • Community-onset without risk factors (i.e. empiric cefepime): ○ **Change** to ampicillin/sulbactam ○ Step-down to PO amoxicillin-clavulanate when clinically stable for 48-72h • Hospital-onset regardless of risk factors OR community-onset with risk factors: ○ If using combination therapy, **stop** tobramycin or levofloxacin ○ Continue empiric beta-lactam to complete a 7-day course **Clinical worsening:** • Obtain a respiratory sample for culture • Consult ID if strongly suspect symptoms are driven by bacterial infection

Combination therapy with two anti-pseudomonal agents is rarely indicated beyond the initial empiric setting. ID consult is recommended if considering continuation beyond 48h for patients without antimicrobial susceptibilities pending.

‡If prior colonization or infection is from an anatomical site outside the respiratory tract (e.g. urinary, intra-abdominal), consider de-escalation to ampicillin/sulbactam or ceftriaxone as above for patients showing early clinical improvement and hemodynamic stability.

Footnotes:

† **CAP clinical signs of instability (if different than patient baseline status)**

- HR ≥ 100 bpm
- RR ≥ 24 breaths/min
- SBP ≤ 90 mmHg
- Arterial O_2 sat ≤ 90% or pO_2 ≤ 60 mmHg on room air
- Altered mental status

* Dose may need to be adjusted for renal dysfunction

***

- See Beta-lactam Allergy Evaluation and Empiric Therapy Guidance document for further information on safe treatment strategies in patients with purported β-lactam allergies.
- Consult the Beta-Lactam Allergy Evaluation Service (BLAES) to assess patients with purported β-lactam allergies
- Low risk includes remote unknown reaction, pruritis without rash, urticaria/hives with no other symptoms, and mild to severe rash with no other symptoms.
- High risk includes:
    1. Anaphylaxis
    2. Cardiovascular symptoms (syncope/passing out, arrhythmia)
    3. Angioedema/swelling
    4. Respiratory distress
- Severe non-IgE mediated PCN allergy includes organ damage (kidney/liver), drug-induced immune-mediated anemia/thrombocytopenia/leukopenia, Stevens-Johnson Syndrome/Toxic Epidermal Necrosis, rash with pustules, DRESS, serum sickness

Last revised: 06/2025	**Antimicrobial Stewardship Subcommittee:** 05/2025	**P&T:** 6/2025

References	
	Citation

Revision History	
Date	**Revision**
08/2020	Revised/added pathways B & C
10/2020	Adjusted ceftriaxone dosing
03/2021	Updated vancomycin dosing & hyperlinks
07/2021	Updated pathway B & C empiric antibiotic sections
09/2021	Updated vancomycin infusion reaction terminology
07/2023	Updated VAP section
07/2024	Updated flowchart, combined pathway A (ICU) and pathway A (non-ICU), revised allergy wording
06/2025	Updated pathways A, B & C, added pathway D, ICU flowchart, and Appendix

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source.